MDX-1 Simultaneous multi-frequency + pulse-induction metal detector

Sheets: 1 Power 2 TX driver 3 RX front end 4 ADC 5 MCU 6 Audio/RF + UI link 7 UI board (separate PCB) 8 Coil. All inter-sheet signals are global labels.

Generated from tools/mdx1_design.py — edit the generator, not the .kicad_sch files. Capacitors drawn at a pin belong to that pin.

Power: charger, gauge, rails, TX boost converters

File: power.kicad_sch

TX driver: 3-level H-bridge, current sense, over-current trip

File: tx_driver.kicad_sch

RX front end: coil input protection, 20x differential preamp, ADC drivers

File: rx_frontend.kicad_sch

ADC: 2x ADS127L01 24-bit 512 kSPS, 2.5 V reference

File: adc.kicad_sch

MCU: STM32H743ZI, clocks, debug, storage

File: mcu.kicad_sch

UI, audio, wireless: memory LCD, keypad, IMU, headphone + speaker amps, Bluetooth (separate PCB)

File: ui_audio_rf.kicad_sch

UI board (separate PCB): memory LCD, 6-key pad + pinpoint + power, LEDs, front light

File: ui_board.kicad_sch

Search-coil assembly (DD 11 in): windings, shield, ID chip

File: coil.kicad_sch

Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX

Open hardware design (generated)

Sheet: /

File: mdx1.kicad_sch

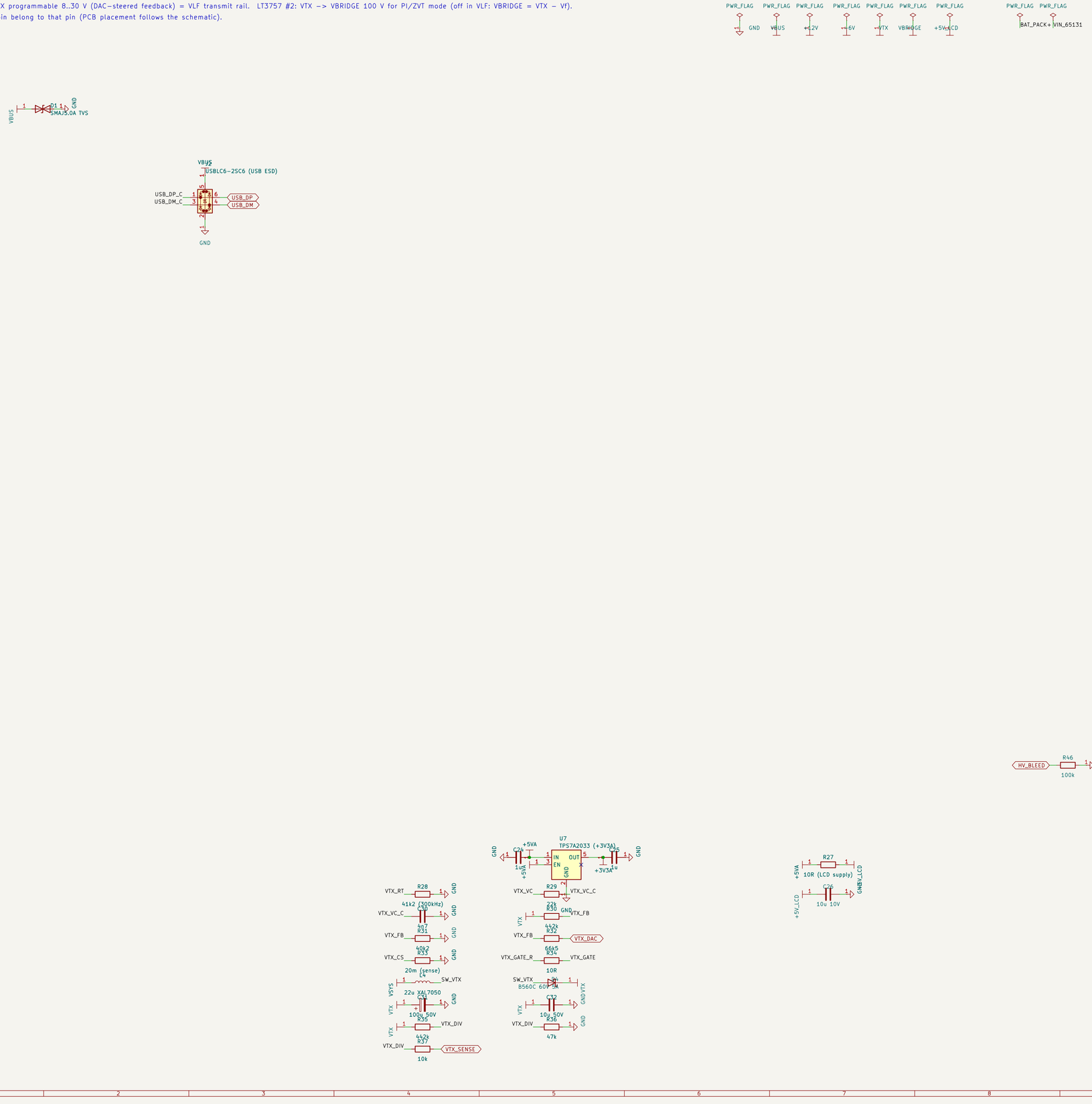
Title: MDX-1 multi-frequency / PI metal detector

Size: A3	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 1/9

MDX-1 POWER TREE

1S Li-ion 21700 (5 Ah) -> BQ24075 power-path charger (USB-C 5 V) -> VSYS. VSYS -> TPS63001 buck-boost +3V3 (digital);
TPS65131 -> +12V (gate drivers, feeds LT3045 +5VA ultra-low-noise LDO) and -6V (feeds LT3094 -5VA LDO); TPS7A20 -> +3V3A (ADC/ref).

LT3757 #1: VSYS -> VTX programmable 8..30 V (DAC-steered feedback) = VLF transmit rail. LT3757 #2: VTX -> VBRIDGE 100 V for PI/ZVT mode (off in VLF: VBRIDGE = VTX - Vf).
Capacitors drawn at a pin belong to that pin (PCB placement follows the schematic).



Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX
Open hardware design (generated)
Sheet: /Power: charger, gauge, rails, TX boost converters/
File: power.kicad_sch

Title: Power: charger, gauge, rails, TX boost converters

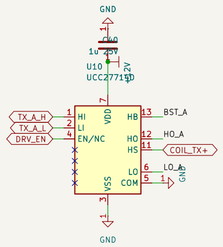
Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 2/9

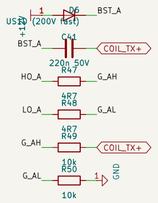
TX H-BRIDGE (150 V MOSFETs, UCC27714 600 V half-bridge drivers, bootstrap high side)

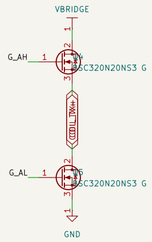
VLF mode: VBRIDGE = 8..30 V, TIM1 CH1/CH1N + CH2/CH2N (dead-time 100 ns) synthesise the optimised 3-level 5-frequency waveform (LUT, 2.44 us edge slots).

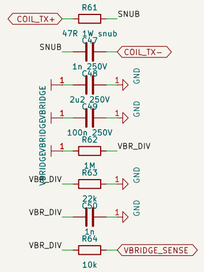
PI/ZVT mode: VBRIDGE = 100 V; fast current reversal through the bridge, zero-voltage hold by low-side freewheel (Q5+Q7 on) with PWM top-up under current feedback.

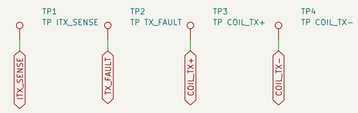
In-line 20 mOhm shunt + INA241A2 (-5..+110 V common mode, 1.1 MHz) gives the TX current reference ITX (0.4 V/A around 2.5 V) for software normalisation VRX/ITX.

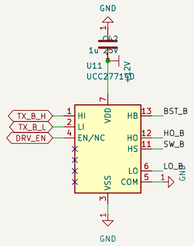


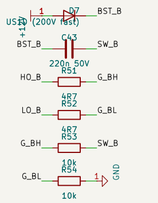


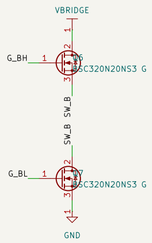




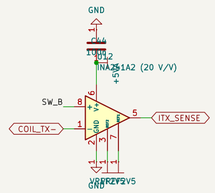


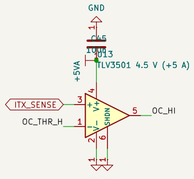


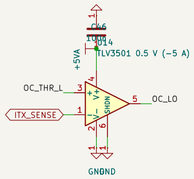


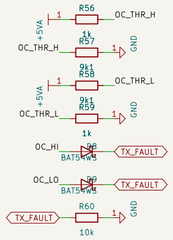












TX_FAULT (active high, wired-OR) -> TIM1 BKIN: hardware shuts all four gates off within one timer clock, independent of firmware.

Gate resistors 4R7 -> ~40 ns edges; dead time set in TIM1 BDTR (100 ns). Bridge decoupling (4u7 + 100n 200 V) sits at the high-side drains.

Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX

Open hardware design (generated)

Sheet: /TX driver: 3-level H-bridge, current sense, over-current trip/

File: tx_driver.kicad_sch

Title: TX driver: 3-level H-bridge, current sense, over-current trip

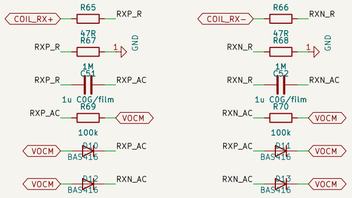
Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 3/9

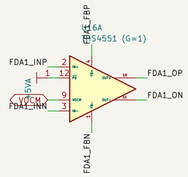
RX FRONT END (OPA1612 in-amp topology, 1.1 nV/rtHz, gain 20; THS4551 fully-differential ADC drivers; VOCM = 1.65 V)

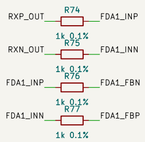
Coil RX winding is AC-coupled and DC-biased to VOCM through 100k. BAS416 clamps to VOCM protect the preamp from PI flyback coupling (recovers < 2 us).
Differential gain = 1 + 2*1k/105R = 20.0 ; common-mode gain 1 (subtracted by the FDA). Input-referred noise ~2.3 nV/rtHz incl. 47R and coil resistance.

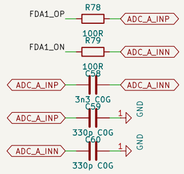
Full scale at ADC (+-2.5 V diff) corresponds to +-125 mV at the coil - the 24-bit ADC gives >110 dB headroom so no AGC is needed in any soil.

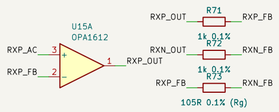




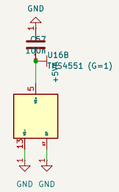


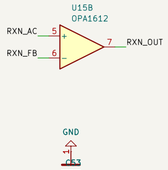


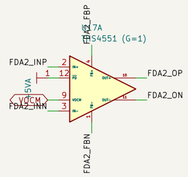


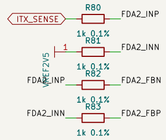


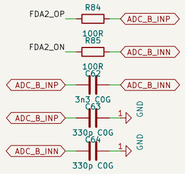


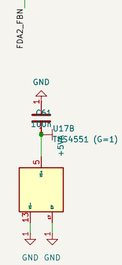




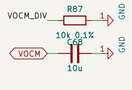






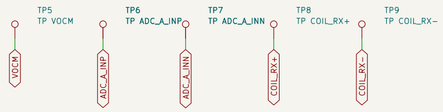












VOCM = 2.5 V * 10k/(10k+5.11k) = 1.654 V = AVDD/2 of the ADC. OPA1656 B is an unused unit (follower on GND).

Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX
Open hardware design (generated)
Sheet: /RX front end: coil input protection, 20x differential preamp, ADC drivers/
File: rx_frontend.kicad_sch

Title: RX front end: coil input protection, 20x differential preamp, ADC driver

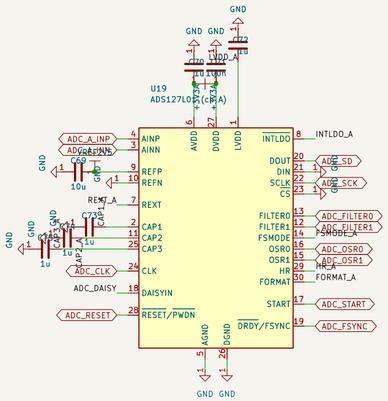
Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 4/9

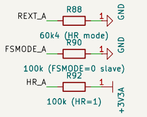
ACQUISITION (2 x ADS127L01, wideband−1 FIR, OSR 32 −> 512 kSPS each, 16.384 MHz clock from MCU MCO2 −> coherent with TX timers)

ADC A digitises the RX coil; ADC B digitises the TX current reference. Identical filters => identical group delay, so the complex ratio VRX/ITX has no phase error.

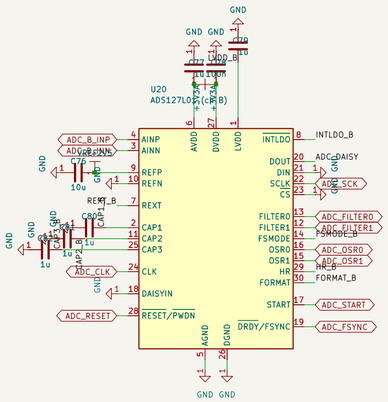
Interface: frame−sync slave mode (FORMAT=1, FSMODE=0), daisy−chained (B DOUT −> A DAISYIN). The MCU SAI1 block B runs TDM: 32.768 MHz SCLK, 512 kHz FSYNC, 2 x 32−bit slots −> one DMA stream, zero CPU.

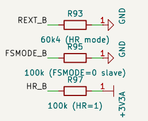
TX period = 100 samples (5.12 kHz fundamental): demodulation is an exact 100−point DFT per period at bins 1,2,3,4,8 (5.12/10.24/15.36/20.48/40.96 kHz).



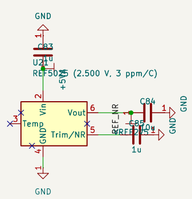


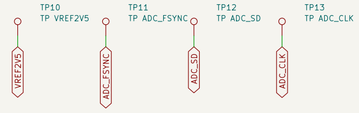














Hardware straps: FILTER[1:0]=00 wideband−1, OSR[1:0]=00 (512 kSPS) are driven by MCU GPIO so firmware can drop to 256/128 kSPS for lower power.

FORMAT=1 (frame−sync), FSMODE=0 (slave): FSYNC (512 kHz) and SCLK (32.768 MHz) come from SAI1_B: A's DOUT carries 48 bits per frame (A then B via DAISYIN). Daisy−chain is supported in frame−sync slave mode (DS 8.5.2.2.6).

Simultaneous 5−frequency VLF−IB + bipolar PI, 24−bit RX

Open hardware design (generated)

Sheet: /ADC: 2x ADS127L01 24−bit 512 kSPS, 2.5 V reference/

File: adc.kicad_sch

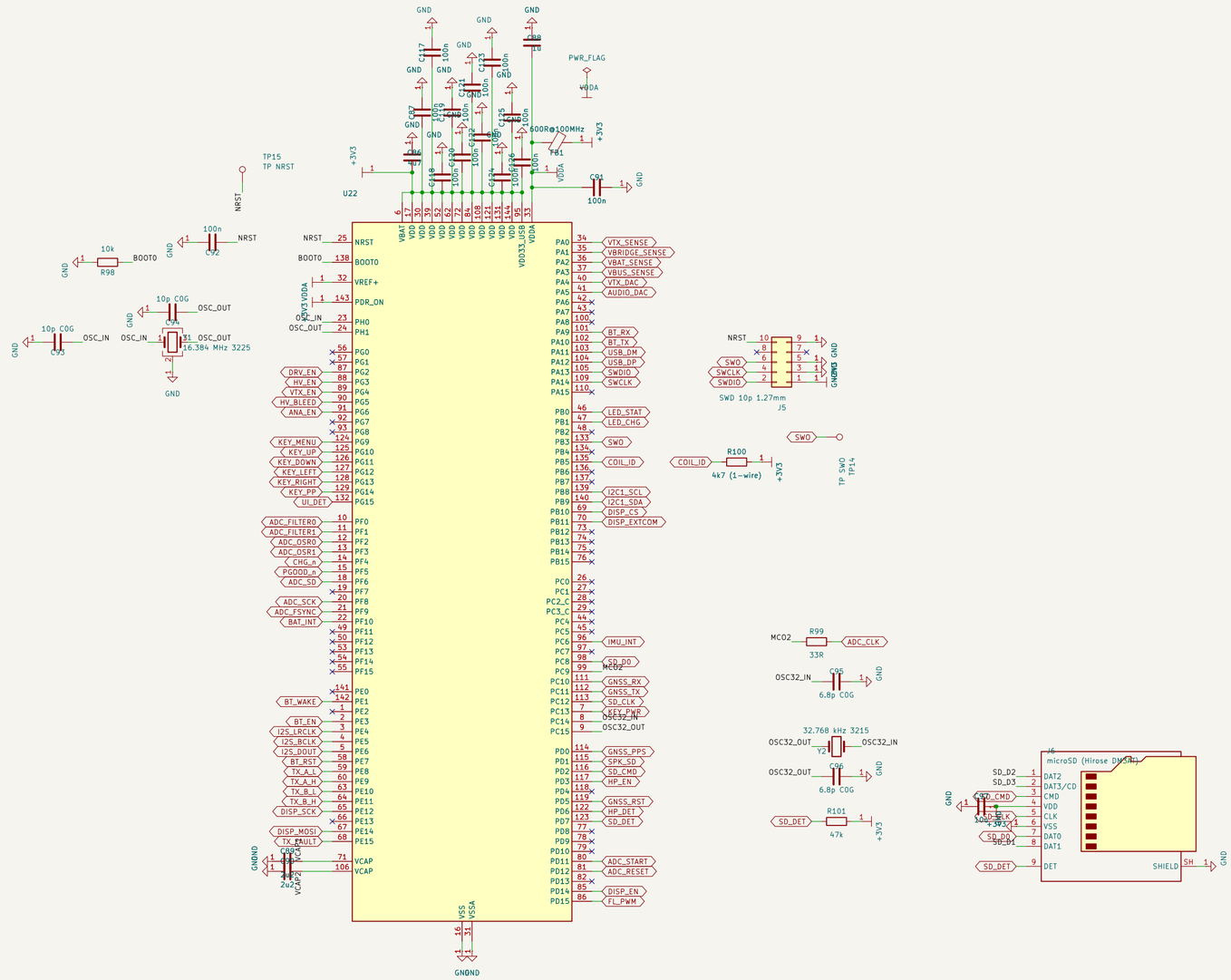
Title: ADC: 2x ADS127L01 24−bit 512 kSPS, 2.5 V reference

Size: A2	Date: 2026−09−05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 5/9

PROCESSOR STM32H743ZIT6 (Cortex-M7 480 MHz, double-precision FPU, 2 MB flash, 1 MB RAM)

HSE 16.384 MHz -> PLL1 (458.752 MHz SYSCLK) and MCO2 = HSE = 16.384 MHz to both ADCs. TIM1 (229.376 MHz) generates the TX waveform: 44800 ticks per 195.3125 us period.
SAI1 block B (PF8 SCK, PF9 FS, PF6 SD) is the ADC data interface: TDM master, 32.768 MHz bit clock from PLL2, 512 kHz frame sync, two 32-bit slots per frame, circular DMA.

All sampling, transmit timing and demodulation are derived from the single crystal -> fully coherent; noise-cancel shifts the TX fundamental by re-scaling the LUT period.



Unused GPIO are left open (ERC no-connect). SD in 1-bit SDMMC mode (PC9 is reserved for MCO2 -> ADC clock). PC13 = WKUP4.

Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX

Open hardware design (generated)

Sheet: /MCU: STM32H743ZI, clocks, debug, storage/
File: mcu.kicad_sch

Title: MCU: STM32H743ZI, clocks, debug, storage

Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 6/9

USER INTERFACE, AUDIO AND RADIO

Display, keys, LEDs and front-light live on a separate UI board (sheet 7) linked by a 20-way 0.5 mm FFC; the front-light MOSFET driver, LSM6DSM IMU, audio and radios stay on the main board.
Audio: DAC -> 7 kHz RC -> TPA6132A2 headphone amp (cap-free) and PAM8302 class-D speaker; I2S to a BM83-class Bluetooth 5 audio module (low-latency codec) for wireless phones.

PWR_FLAG
HPVSS

+5V_LCD
C98 1u GND

J7
UI link FFC 20p 0.5mm
GND
DISP_SCK
DISP_MOSI
DISP_CS
DISP_EXTCOM
DISP_EN
UI_DET
LED_STAT
LED_CHG
KEY_MENU
KEY_UP
KEY_DOWN
KEY_LEFT
KEY_RIGHT
KEY_PP
KEY_PWR
FL_K

UI_DET
R200 10k
+3V3

AUDIO_DAC
R109
AUDIO_FLT
C101 4n7 AUDIO_FLT GND
C102 1u AUDIO_FLT AUDIO_IN
C103 1u HP_INN GND
C104 1u AUDIO_FLT SPK_INP
C105 220n SPK_INN GND

GND
GNDC107 2n2
C106 2n2
+3V3
U24
TPA6132A2 (HP amp)
AUDIO_IN
HP_INN
AUDIO_IN
HP_INN
HP_EN
INR+
INR−
INL+
INL−
G0
G1
EN
VDD
HPVDD
SGND
PGND
HPVSS
EP
OUTR
OUTL
CPP
CPN
HP_R
HP_L
CPP
CPN
GNDC108 1u
GND

CPP C109 1u CPN

HP_DET
R110 100k
+3V3

J8
3.5 mm jack
GND
S
R HP_R
T HP_L
TN HP_DET

Key inputs use the MCU pull-ups; keys switch to GND on the UI board. FL_K is the front-light cathode, switched here by Q_FL (PWM).

FL_K
R105 22R FL_R
FL_PWM
R106 100R FL_G
FL_G
R107 100k GND
FL_R
FL_G
Q2 AO7002
3
1
2
GND

GND
C110 1u
U25
PAM8302B (2.5 W class-D)
VSPK
SPK_INP
SPK_INN
SPK_SD
IN+
IN−
SD
VDD
GND
OUT+
OUT−
SPK+
SPK−

J9
Speaker 8R
SPK+
SPK−

VSYS
C111 10u GND

GND
C112 1u
U23
LSM6DSM (IMU)
+3V3
+3V3
IMU_SA0
I2C1_SDA
I2C1_SCL
+3V3
SDO_Aux
SDO/SA0
SDX
SCX
OCS_Aux
SDA
SCL
CS
VDDIO
VDD
GND
INT1
INT2
IMU_INT

IMU_SA0
R108 0R (addr) GND

GND
GNDC113 1u
C114 1u
U27
MAX-M103 (GNSS)
+3V3
GNSS_RX
GNSS_TX
RXD
TXD
SDA
SCL
EXTINT
SAFEBOOT
RESET
VIO_SEL
VCC
VCC_IO
V_BCKP
VCC_RF
LNA_EN
RF_IN
TIMEPULSE
GND
GNSS_RF
GNSS_PPS

GNSS_RF
R111 0R (match, series) GNSS_ANT
GNSS_ANT
C115 DNP (match, shunt) GND

GNSS_ANT
U28
GNSS chip antenna
GND

U26
BM83SM1-00AA
I2S_DOUT
I2S_LRCLK
I2S_BCLK
DR1
RFS1
SCLK1
DT1
MCLK1
AOHPR
AOHPM
AOHPL
MICN2
MICP2
AIR
AIL
MICN1
MICP1
MICBIAS
GND
DMIC_CLK
MIDC1_R
MIDC1_L
P3_2
P2_6
ADAP_IN
BAT_IN
SYS_PWR
VDD_IO
PWR(MFB)
BT_EN
GND
VBAT
C117 10u
GNDGNDGND
GND
GND
GND
P0_0/UART_TX_IN
P3_7/UART_CTS
P1_3/TCK_CPU/SDA
P1_2/TDI_CPU/SCL
P0_7
P0_1
RST_N
P2_3
P1_6/PWRM1
P0_5
P2_7
P0_3
DP
DM
P0_6
LED2
P0_2
LED1
P3_4/UART_RTS
P8_5/UART_TXD
P8_6/UART_RXD
SK2_KEY_AD
SK1_AMB_DET
BT_WAKE
BT_RST
BT_TX
BT_RX

Antenna: 50 ohm trace, series 0R and shunt DNP pads for the matching network the antenna vendor recommends for this ground-plane size.

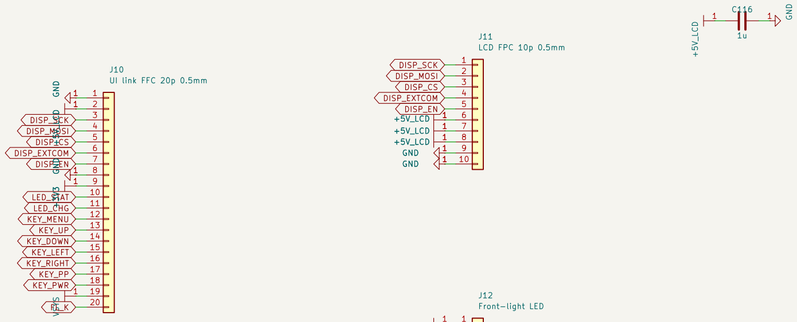
Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX
Open hardware design (generated)
Sheet: /UI, audio, wireless: memory LCD, keypad, IMU, headphone + speaker amps, Bluetooth audio, G
File: ui_audio_rf.kicad_sch

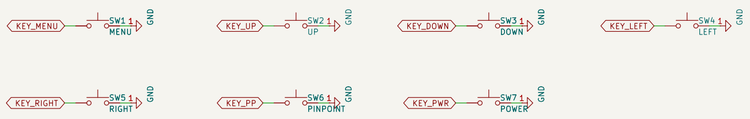
Title: UI, audio, wireless: memory LCD, keypad, IMU, headphone + speaker ar

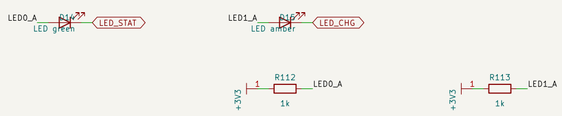
Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 7/9

UI BOARD (separate 2-layer PCB behind the display window; 20-way FFC to the main board)

Sharp LS027B7DH01 memory LCD (400x240, 5 V panel, 3.3 V logic). Keys: MENU (short = menu/enter, hold 0.8 s = back), UP/DOWN, LEFT/RIGHT, PINPOINT trigger (hold 1.5 s = ground grab), POWER (hold 1.5 s = off, wakes the MCU).
Every net on this sheet is a global label shared with the UI link connector on sheet 6. FFC: 20-way 0.5 mm, same-side contacts (type A), pin 1 to pin 1.







PINPOINT is a trigger-style switch under the handle; the other six are the front keypad. Four M2.5 holes; the LCD is bonded to the board with its FPC folded into J_LCD.

Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX
Open hardware design (generated)

Sheet: /UI board (separate PCB): memory LCD, 6-key pad + pinpoint + power, LEDs, front-light/		
File: ui_board.kicad_sch		
Title: UI board (separate PCB): memory LCD, 6-key pad + pinpoint + power,		
Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 8/9

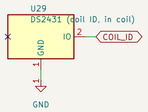
SEARCH COIL (not on the main PCB — potted DD coil on 1 m shielded cable)

TX winding: 0.36 mH, 0.63 ohm, 23 turns 0.8 mm Cu-equivalent Litz. RX winding: 3.05 mH, 17.6 ohm, 63 turns 0.25 mm, SRF > 150 kHz incl. cable.
Windings wrapped with graphite (200–2000 ohm/sq) electrostatic shield with a gap, drained to cable shield. Null overlap 20.8 mm; residual < 0.3 ohm at 41 kHz after trimming.

DS2431 1-Wire EEPROM stores coil type, null phase/amplitude per frequency and factory calibration; read at power-up and on hot-swap.







Geometry per the coil manual: TX D left, RX D right, symmetric overlap 20.8 mm; foam-core, epoxy potted, ABS shell, IP68; M12 8-pin A-coded plug.

Simultaneous 5-frequency VLF-IB + bipolar PI, 24-bit RX
Open hardware design (generated)

Sheet: /Search coil assembly (DD 11 in): windings, shield, ID chip/		
File: coil.kicad_sch		
Title: Search coil assembly (DD 11 in): windings, shield, ID chip		
Size: A2	Date: 2026-09-05	**Rev: A4**
KiCad E.D.A. 10.0.5		Id: 9/9